MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Skin and Soft Tissue Infections

Clinical Setting	Empiric Therapy	Duration	Comments
Minor Skin Infections Impetigo • Secondarily infected skin lesions such eczema, ulcers, or lacerations	Mupirocin 2% topical ointment BID	7 days	
Abscess, Furuncles, and Carbuncles **Abscesses** - collections of pus within the dermis and deeper skin tissues **Furuncle** - infection of the hair follicle in which purulent material extends through the dermis into the subcutaneous tissue, where a small abscess forms **Carbuncle** - coalescence of several furuncles into a single inflammatory mass	INCISION AND DRAINAGE (I&D) IS RECOMMENDED AS PRIMARY MANAGEMENT, AND ANTIBIOTICS ARE NOT INDICATED UNLESS PATIENT MEETS ONE OF THE FOLLOWING CRITERIA: • Severe, extensive, rapidly progressive cellulitis, or abscess > 2 cm • Signs or symptoms of systemic illness • Elderly, immunosuppressed, active neoplasm or diabetes mellitus • Circumstances where abscess is difficult to drain • Associated septic phlebitis • Inadequate response to I&D alone EMPIRIC ORAL ANTIBIOTIC THERAPY OR ORAL STEP-DOWN THERAPY MEETING ABOVE CRITERIA: *Preferred:* **TMP-SMX*** 1-2 DS tabs PO BID Alternative for *receiving potassium sparing diuretics, chronic kidney disease, or sulfa allergy:* **Doxycycline** 200 mg PO x1, then 100 mg PO BID *Alternative for patients with extensive or severe disease, OR I&D and oral antibiotics were ineffective:* **Linezolid** 600 mg PO BID EMPIRIC IV ANTIBIOTIC THERAPY FOR HOSPITALIZED PATIENTS: *Preferred:* **Vancomycin*** IV (see nomogram)	5-7 days	• Close clinical follow-up is recommended, especially in patients no receiving antibiotic therapy • Cultures and susceptibility are recommended when I&D is performed • Renal dose adjustment may be required for vancomycin and trimethoprim-sulfamethoxazole • *MRSA* resistance rates are lowest for TMP-SMX (9%) and doxycycline (13%), compared to clindamycin (34%). • Empiric therapy should target MRSA until susceptibilities are known, and then therapy should be tailored. For patients with MSSA, preferred oral step-down therapy is cephalexin, TMP-SMX, or doxycycline if patient has severe beta-lactam allergy • Bactrim should be used with caution in patients with renal dysfunction and risk for hyperkalemia

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Non-Purulent Cellulitis (Absence of purulent drainage or exudate, ulceration, and no associated abscess) Empiric therapy for β-hemolytic streptococcus is recommended. If there is a concern for **necrotizing fasciitis**, please see treatment recommendations listed under that section	EMPIRIC IV ANTIBIOTIC THERAPY FOR HOSPITALIZED PATIENTS, INCLUDING PATIENTS LOW-TO-HIGH RISK PCN ALLERGY***: *Preferred:* **Cefazolin***2 g IV q8h EMPIRIC ORAL ANTIBIOTIC THERAPY FOR OUTPATIENT THERAPY, OR ORAL STEP-DOWN THERAPY: *Preferred* **Cephalexin*** 1000 mg PO TID *Alternative:* **TMP-SMX** 1-2 DS tablets BID **Linezolid** 600 mg PO BID *Alternative for patients at risk for MRSA non-purulent cellulitis:* **Vancomycin*** IV (see nomogram) if MRSA coverage is indicated Patients at risk for MRSA: • Cellulitis worse on > 48 hours of IV β-lactam therapy • Known MRSA colonization • Prior history of MRSA infection • Recent intravenous drug use • Severe sepsis or septic shock EMPIRIC ORAL ANTIBIOTIC THERAPY FOR OUTPATIENT THERAPY, OR ORAL STEP-DOWN THERAPY: *Preferred* **TMP-SMX*** 1-2 DS BID to cephalexin, if patient presents with risk factors for MRSA (listed above) ALTERNATIVE FOR PATIENTS WITH ALLERGY OR INTOLERANCE TO TMP-SMX (IN PATIENTS WITH OR WITHOUT RISK FOR MRSA): **Linezolid** 600 mg PO BID	5 days for patients with rapid clinical response. Longer duration of therapy is indicated if infection has not improved	• Blood cultures and cultures from purulent SSTI are recommended for patients with fever, rapidly progressive cellulitis, or signs of systemic illness • Adjust antimicrobial therapy based on culture results • Consider initial aggressive dosing of antibiotics if severe sepsis or morbidly obese patient • Group B Strep is intrinsically resistant to doxycycline and should be avoided in non-purulent cellulitis

Clinical Setting	Empiric Therapy	Duration	Comments
Purulent Cellulitis (Purulent drainage or exudate without a drainable abscess) Empiric therapy for CA-MRSA is recommended Therapy for β-hemolytic streptococci is likely to be unnecessary	EMPIRIC ANTIBIOTIC THERAPY FOR HOSPITALIZED PATIENTS: *Preferred:* **Vancomycin*** IV (see nomogram) *Alternative for vancomycin, if documented allergy or intolerance:* **Linezolid**** 600 mg PO BID EMPIRIC ORAL ANTIBIOTIC THERAPY, OR ORAL STEP-DOWN THERAPY FROM VANCOMYCIN **TMP-SMX*** 1-2 DS tabs PO BID OR **Doxycycline** 200 mg x1, then 100 mg PO BID OR **Linezolid** 600 mg PO BID	5-10 days Longer duration of therapy is indicated if infection has not improved	• Blood cultures and cultures from purulent SSTI are recommended for patients with fever, rapidly progressive cellulitis, or signs of systemic illness • MRSA resistance rates are lowest for TMP-SMX (9%) and doxycycline (13%), compared to clindamycin (34%). • Empiric therapy should target MRSA until susceptibilities are known, and then therapy should be tailored. For patients with MSSA, preferred oral step-down therapy is cephalexin, or TMP-SMX or doxycycline if patient has severe beta-lactam allergy

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Necrotizing Fasciitis, including Fournier's Gangrene Early and aggressive surgical exploration and debridement is critical Emergent surgical consultation is recommended	PREFERRED: **Piperacillin-tazobactam*** 4.5 g IV q6h + **Linezolid** 600 mg IV/PO q12h ALTERNATIVE FOR PATIENTS WITH LOW-TO-HIGH RISK PENICILLIN ALLERGY, INCLUDING ANAPHYLAXIS***: **Cefepime*** 2 g IV q8h **+ Linezolid** 600 mg IV/PO q12h **+ metronidazole** 500 mg q12h Alternative for patients with cephalosporin allergy, and for patients with known colonization or prior infection with ESBL organism: **Meropenem** 2 g IV q8h + **Linezolid** 600 mg IV/PO q12h *Alternative for patients with severe non-IGE beta-lactam allergy***:* **Aztreonam*** 2 g IV q8h **+ Linezolid** 600 mg IV/PO q 12h **+ metronidazole** 500 mg IV/PO q12h	Optimal duration is depending on source control and clinical response. Please consult Infectious Diseases.	• Emergent surgical and infectious disease consultation is recommended • Linezolid is initiated for anti-toxin activity for \streptococcal and staphylococcal infections. Bactrim should be used with caution in patients with renal dysfunction and risk for hyperkalemia • Consider IVIG therapy for Group A Strep toxic shock syndrome in consultation with Infectious Diseases Consult

Clinical Setting	Empiric Therapy	Duration	Comments
Superficial Surgical Site Infections Infections involving the subcutaneous tissue within 30 days of operation For SSI involving deep tissue or organ space or complicated by sepsis/septic shock, see below or organ specific guidelines (Intra-abdominal, Gynecology, Meningitis, Endocarditis, Bone and Joint) Suture removal plus incision and drainage should be performed If there is a concern for necrotizing fasciitis, please refer to that section	**All systemic antimicrobial therapy should be used in combination with opening the incision and evacuation of infected material** PREFERRED EMPIRIC INPATIENT THERAPY FOR SUPERFICIAL SSI, WITHOUT RISK FOR MRSA OR GNR (SEE BELOW): **Cefazolin*** 2 g IV q8h OR **Cephalexin** 1000 mg PO TID for mild-moderate infection, or oral step-down therapy ALTERNATIVE EMPIRIC INPATIENT THERAPY FOR PATIENTS WITH HIGH RISK OF MRSA: **Vancomycin*** IV (see nomogram) OR **Linezolid** 600 mg IV/PO q 12h RISK FACTORS FOR GNR: **Cefuroxime*** 1.5 g IV q8h + **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram), if risk for MRSA **Oral Alternative** for mild-moderate infection, or oral step-down therapy: **Amoxicillin-clavulanate** 875 mg PO BID *Alternative for patients with severe non-IGE penicillin allergy***:* **Aztreonam*** 2 g IV q8h + **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram)	5 -7 days Therapy may need to be extended based on severity of infection and response to treatment	• Risk factors for GNR include: ○ Operation of axilla, GI tract, perineum or female genital tract • Risk factors for MRSA include: ○ nasal colonization ○ prior MRSA infection ○ recent hospitalization ○ recent antibiotics • Antibiotics should be adjusted based on Gram stain, cultures and sensitivities obtained from I&D • Wound infection and systemic illness in the first 4 days (especially the first 48 hours) should prompt close wound examination for evidence of streptococcal or clostridial necrotizing infection. See necrotizing fasciitis section if concern exists.

Clinical Setting	Empiric Therapy	Duration	Comments
Deep tissue Surgical Site Infections or any SSI complicated by sepsis/septic shock Infections involving the deep fascia or muscle within 30 days of operation For SSI with organ space involvement, see specific guidelines for Intra-abdominal, Gynecologic, Meningitis, Endocarditis, and Bone and Joint for specific recommendations For necrotizing infections, see Necrotizing Fasciitis section	**All systemic antimicrobial therapy should be used in combination with opening the incision and evacuation of infected material** EMPIRIC INPATIENT THERAPY FOR DEEP TISSUE SSI OR COMPLICATED BY SEPSIS/SEPTIC SHOCK: **Piperacillin-tazobactam*** 4.5 g IV q6h + **Vancomycin*** IV (see nomogram) ALTERNATIVE WITH LOW-TO-HIGH RISK PCN ALLERGY***: **Cefepime*** 2 g IV q8h + **Vancomycin*** IV (see nomogram) + **metronidazole** 500 mg IV/PO q12h Alternative for patients with cephalosporin allergy: **Meropenem** 2 g IV q8h + **Vancomycin** * IV (see nomogram) *Alternative for patients with severe non-IGE beta-lactam allergy****: **Aztreonam*** 2 g IV q8h + **Vancomycin*** IV (see nomogram) + **metronidazole** 500 mg IV/PO q12h	7-10 days Therapy may need to be extended based on severity of infection and response to treatment	

Clinical Setting	Empiric Therapy	Duration	Comments
Traumatic Wound Infections of Extremity Usually polymicrobial from environmental contamination If there is a concern for necrotizing fasciitis, please refer to that section Please see separate guidelines for bite wounds or for wounds complicated by osteomyelitis or open fracture	PREFERRED INPATIENT THERAPY FOR HEMODYNAMICALLY STABLE PATIENTS WITH ACUTE INFECTION (< 5 DAYS FROM INJURY), AND NO RISK FOR MRSA: **Ampicillin-sulbactam*** 3 g IV q6h ALTERNATIVES FOR LOW-TO-HIGH RISK PCN ALLERGY***:: **Cefazolin*** 2 g IV q8h + **Metronidazole** 500 mg PO/IV q12h PREFERRED INPATIENT THERAPY FOR PATIENTS WITH HIGH RISK OF MRSA: **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram) PREFERRED THERAPY IN PATIENTS WITH ANY OF THE FOLLOWING: • Sepsis and traumatic wound infection • Development of infection > 5 days after injury • Significant water exposure and consult infectious diseases team **Piperacillin-tazobactam*** 4.5 g IV q6h + **Vancomycin*** IV (see nomogram) ALTERNATIVE FOR PATIENTS WITH LOW-TO-HIGH RISK PCN ALLERGY***: **Cefepime** *2 g IV q8h + **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram) Alternative for patients with cephalosporin allergy: **Meropenem** 2 g IV q8h + **Vancomycin** * IV (see nomogram) *Alternative for patients with severe non-IGE penicillin allergy***:* **Aztreonam*** 2 g IV q8h + **Metronidazole** 500 mg PO/IV q8h + **Vancomycin*** IV (see nomogram)	7-14 days Therapy may need to be extended based on severity of infection and response to treatment	• **Risk factors for MRSA include**: ○ nasal colonization ○ prior MRSA infection ○ recent hospitalization ○ recent antibiotics • Traumatic wounds without evidence of local infection or systemic signs of infection typically do not need antimicrobial therapy beyond appropriate surgical prophylaxis • Empiric therapy should take into account site of wound, prior cultures and colonization • Debridement of devitalized tissues and contaminating debris critical to source control and successful healing • Antibiotic therapy should be tailored to results of deep tissue Gram stain, culture and sensitivities • Consider initial aggressive dosing of antibiotics if severe sepsis or morbidly obese patient • Osteomyelitis would require 6-8 weeks of therapy- please see osteomyelitis guidelines

Clinical Setting	Empiric Therapy	Duration	Comments
Complicated SSTI without Osteomyelitis Decubitus or sacral wound infection	**Obtain deep tissue culture prior to starting antibiotic therapy if patient is hemodynamically stable** PREFERRED EMPIRIC INPATIENT THERAPY: **Vancomycin*** IV (see nomogram) + **Piperacillin-tazobactam*** 4.5 g IV q6h (when gram-negative organisms are suspected) ALTERNATIVE LOW-TO-HIGH RISK PENICILLIN ALLERGY: **Cefepime** *2 g IV q8h + **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram) Alternative for patients with high-risk cephalosporin allergy: **Meropenem** 2 g IV q8h + Vancomycin * IV (see nomogram) *Alternative for patients with severe non-IGE beta-lactam allergy***:* **Aztreonam*** 2 g IV q8h + **Metronidazole** 500 mg PO/IV q12h + **Vancomycin*** IV (see nomogram)	7- 14 days Duration may be longer in the setting of poor clinical response	• Blood cultures are recommended • When clinically indicated, evaluate for underlying osteomyelitis • Cultures of purulent drainage or deep tissue cultures when feasible are recommended • Tailor therapy according to culture data • Consider ID consult • Surgical debridement is an important component in management • Osteomyelitis with decubitis or sacral osteo please see osteomyelitis guidelines

*Adjustment in dose is necessary with renal insufficiency
**Complete information regarding drug interactions with linezolid

- See Beta-lactam Allergy Evaluation and Empiric Therapy Guidance document for further information on safe treatment strategies in patients with purported β-lactam allergies.
- Consult the Beta-Lactam Allergy Evaluation Service (BLAES) to assess patients with purported β-lactam allergies
- Low risk includes remote unknown reaction, pruritis without rash, urticaria/hives with no other symptoms, and mild to severe rash with no other symptoms.
- High risk includes:
    1. Anaphylaxis
    2. Cardiovascular symptoms (syncope/passing out, arrhythmia)
    3. Angioedema/swelling
    4. Respiratory distress
- Severe non-IgE mediated PCN allergy includes organ damage (kidney/liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, Stevens-Johnson Syndrome/Toxic Epidermal Necrosis, rash with pustules, DRESS, serum sickness

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References		
	Citation	

1	Stevens DL, Bisno AL, ChamberHF, et al. Practice Guidelines for the Diagnosis and Management of Skin and Soft Tissue Infections: 2014 Update by the Infectious Diseases Society of America. Clinical Infectious Diseases 2014;59(2):e10-52.
2	Lipsky BA, Berendt RA, Cornia PB, et al. 2012 Infectious Diseases Society of America Clinical Practice Guideline for the Diagnosis and Treatment of Diabetic Foot Infections. Clinical Infectious Diseases 2012;54(12):132–173.
3	Liu C, Bayer A, Cosgrove SE, et al. Clinical Practice Guidelines by the Infectious Diseases Society of America for the Treatment of Methicillin-Resistant *Staphylococcus Aureus* Infections in Adults and Children. Clin Infect Dis 2011:52;1-38.

Revision History

Date	Revision
03/2021	Adjusted vancomycin dosing & hyperlinks
09/2021	Updated vancomycin infusion reaction terminology
09/2024	Added linezolid options for alternative treatments,

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source

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